JOINT SESSION · WEEK 5

Mid-point clinic

Comparing Europe with Data. Check how every team is doing and solve problems with the methods before deliverable 2 (week 6).

Week 5 · 16 Nov – 22 Nov 2026 · 45 minutes · Microsoft Teams

Ulysseus EUROPEAN UNIVERSITY

UNIVERSIDAD DE SEVILLA

TECHNICAL UNIVERSITY OF KOŠICE

UCG Univerzitet Crne Gore

AGENDA · 45 MINUTES

What we will do today



20 min	Lightning updates: each team presents its slide in 2 minutes.
20 min	Methods clinic in break-out rooms: regression, classification, PCA and clustering. Teachers rotate between teams.
5 min	Wrap-up: what to submit in deliverable 2.

LIGHTNING UPDATES · 2 MINUTES PER TEAM

Bring one slide with four boxes

Question

Your research question in one sentence.

Data

Indicators, countries and years you use.

First result

One chart or one number, with one sentence of interpretation.

Main problem

The one thing you want help with today.

Rules

- One slide only. The template is in the student templates file on the website.
- One team member presents; 2 minutes, timed.
- Be honest about the problem: that is what the clinic is for.

WHICH METHOD?

Start from your question, not from the method

If your question is…	Use…	Example from the COIL
Do two indicators move together?	Correlation and a scatter plot	Internet use and income (r = 0.73)
How much does Y change when X changes, other things equal?	Linear regression	Youth unemployment, education and income
Does the relation hold within countries over time?	Panel regression with country fixed effects	Ageing and health expenditure, 2010–2023
Which group does a country belong to (high/low)?	Classification (logistic regression)	High or low female participation
Can many indicators be summarised in a few?	Principal component analysis (PCA)	Six indicators → development and ageing
Which countries look alike?	Clustering (k-means)	Digital profiles of European countries

Each method is explained step by step in the methods module on the website: menus first, code optional.

CLINIC

Problems we see every year, and what to do

"We only have about 30 observations."

Use 2–3 explanatory variables at most, report n, or move to the panel (countries × years).

"The coefficient has the wrong sign."

Look at the scatter plot first: outliers, units, and variables you left out.

"p = 0.08: is it significant?"

Report it as weak evidence, with the confidence interval. Do not try models until one works.

"Montenegro has no data."

Apply the missing-values rules: change year or indicator, leave out, interpolate, or another source.

"We cannot name the components."

Read the loadings above 0.4 in absolute value. The sign of a component is arbitrary.

"k-means gives other groups each time."

Standardise, fix the seed, use many starts (n_init / nstart = 25).

"Countries have different years."

Use the same year for all, or the average of three years.

"Our three countries do not fit."

That is a finding. Explain it with the national context.

DUE SUNDAY 29 NOVEMBER, 23:59 CET

Deliverable 2 · Analysis report

It must include

- Data: the dataset you analysed and a data dictionary (name, code, definition, unit, source and years of every variable), with missing values and how you handled them.
- Exploratory analysis: descriptive statistics and at least three charts, one comparing Spain, Slovakia and Montenegro with the rest of Europe.
- Regression or classification: model specification, results table and interpretation in plain English.
- PCA and clustering: variance explained, number of clusters and why, and where Spain, Slovakia and Montenegro fall.
- Limitations (sample size, missing data, causality) and your three main conclusions.
- A short note on your use of AI tools (which tool, for what, and how you checked the result), or a sentence saying you did not use any.

Format

Report made with the Word template, saved as PDF + your analysis file (Python or R notebook, Gretl script or Power BI file). Add your dataset only if you used data from outside the COIL dataset. Submit it with the Submit button on the website.

WRAP-UP

From now to the final presentation

Week 5

Regression and classification

Your regression or classification model is estimated and interpreted.

Week 6

Dimensionality reduction and clustering

Deliverable 2 due Sunday 29 November

Week 7

Dashboard, feedback and rehearsal

Final dashboard and slides ready after feedback from another team; the presentation has been rehearsed.

Week 8

Joint online presentation and reflection

Deliverable 3 due Sunday 13 December